CineRec: A practical multi-modal recommender study with cross-dataset validation and published-baseline comparison

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Author contribution statement. Yulin ZHAO: proposed the research question and designed the study, including the model suite, the content-feature pipeline and the evaluation protocol; implemented the code and carried out all experiments; collected, cleaned and analysed the data; drafted the manuscript and revised and approved the final version.

Abstract

Objective: Many comparative studies in recommender systems are evaluated on a single dataset and lack statistical validation against published baselines. This study asks whether model rankings and the contribution of content features transfer across domains.

Methods: We implement six recommenders of increasing complexity (UserCF, ItemCF, SVD, NeuMF, LightGCN and MultiModalNCF), reproduce two published baselines (BPR-MF and VBPR) under the same leave-last-5-out protocol, and evaluate them on MovieLens 100K and an Amazon Video_Games slice. Neural models are reported as five-seed mean±standard deviation, together with modality-ablation and cold-start studies.

Results: The ladder ordering (UserCF < ItemCF < SVD) and the neural band's lead over SVD hold on both domains; MultiModalNCF is statistically tied with VBPR. Removing all content costs about one seed standard deviation on MovieLens and nothing measurable on Amazon, whereas in the cold-start study the content model clearly outranks the content-blind model.

Limitations: Conclusions rest on a single data split without formal significance testing, epoch budgets are not equalised across models, and the Amazon slice covers heavy users only.

Conclusions: A simple concatenation-based multi-modal extension of NeuMF is competitive with a published content-aware baseline, and content features matter mainly for cold-start rather than for overall ranking quality. The full codebase and evaluation scripts are released for reproduction.

Keywords: Recommender systems; Multi-modal recommendation; Cold-start; Cross-dataset validation; Reproducibility

1. Introduction

Recommender systems are a core component of modern information retrieval, powering personalised content discovery across e-commerce, media and social platforms. The field has seen a rapid evolution of methods, from early collaborative filtering approaches based on user–item interaction matrices to modern deep learning models that incorporate rich side information such as text descriptions, images and knowledge graphs.

Despite this rapid progress, many comparative studies in recommender systems suffer from three limitations: (1) they are evaluated on a single dataset, limiting the generalisability of conclusions; (2) they lack comparison against published baselines under matched conditions; and (3) results are often reported as single-point estimates without statistical significance testing.

In this work we present CineRec, a practical study designed to address these gaps. Our contributions are as follows:

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1. We implement and evaluate six recommendation models spanning the full evolution from classical neighbourhood methods to deep multi-modal models, under a unified evaluation protocol.
2. We reproduce two published baselines—BPR-MF [1] and VBPR [2]—on the same split, allowing direct comparison against established methods.
3. We validate all conclusions across two datasets, MovieLens 100K and Amazon Video_Games, testing whether observed patterns generalise across domains.
4. We report all neural model results as mean±standard deviation over five random seeds, distinguishing real effects from seed noise.
5. We conduct both a modality ablation and a cold-start study, quantifying the contribution of each content channel.

The study design is summarised in Figure 1. The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes our methodology, datasets and evaluation protocol. Section 4 presents the experimental results. Section 5 discusses the findings and their limitations. Section 6 concludes.

2. Literature review

2.1 Neighbourhood-based collaborative filtering

User-based and item-based collaborative filtering [3] are among the earliest and simplest recommendation approaches. They rely on similarity measures between users or items to generate recommendations. While simple, they remain competitive baselines and serve as the foundation for more complex methods.

2.2 Matrix factorisation

Matrix factorisation methods such as SVD [4] and BPR [1] transform the recommendation problem into a latent factor space. BPR, in particular, introduced a pairwise ranking loss that has become a standard for implicit feedback recommendation.

2.3 Neural collaborative filtering

NeuMF [5] combined matrix factorisation with multi-layer perceptrons to model non-linear user–item interactions, demonstrating that deep learning can improve recommendation quality over traditional matrix factorisation.

2.4 Graph-based recommendation

LightGCN [6] simplified graph convolutional networks for recommendation by removing feature transformation and non-linear activation, achieving strong performance with increased efficiency.

2.5 Multi-modal recommendation

VBPR [2] extended BPR with visual features, demonstrating that content information can improve recommendation accuracy, especially for cold-start items. Multi-modal recommendation has since become an active research area, incorporating text, images and other side information.

3. Methodology (Design/Approach)

Our objective is to test whether conclusions about model quality and the value of content features survive two changes that single-dataset studies cannot make: a second domain, and multi-seed repetition against published baselines. The methodology below describes the model suite, the shared content features, the two datasets and the evaluation protocol; all configuration parameters are listed in Table 1.

3.1 Datasets

We conduct experiments on two datasets:

- **MovieLens 100K** [7]: 943 users, 1,682 items and 100,000 ratings (1–5 scale), restricted to users with ≥ 20 ratings.
- **Amazon Video_Games** [8]: 2,288 users, 4,753 items and 75,980 ratings (1–5 scale), restricted to heavy users with ≥ 20 ratings.

The Amazon slice is built deterministically from the 5-core Amazon Reviews 2023 Video_Games dataset by keeping users with ≥ 20 ratings and re-applying the item 5-core rule. The original 5-core dataset is publicly available from the Amazon Reviews dataset repository; our slice definition and processing scripts are included in the codebase.

Poster and overview metadata are retrieved from the TMDB API and remain subject to TMDB's terms; MovieLens 100K is used under its non-commercial research terms, and the project's code is released under the MIT licence.

3.2 Content features

We precompute content features shared across all content-aware models:

- **Text**: Sentence-BERT (all-MiniLM-L6-v2) [9] embeddings of movie overviews, 384-dimensional.
- **Image**: ResNet-50 (pool5) [10] embeddings of movie posters, 2,048-dimensional.
- **Genre**: multi-hot genre vectors, 18-dimensional.

3.3 Model suite

We evaluate six models of increasing complexity plus two published baselines. Table 1 summarises all model configurations, and Figure 2 details the MultiModalNCF architecture. Key details:

- **NeuMF**: neural matrix factorisation combining GMF and MLP branches.
- **LightGCN**: simplified graph convolutional network with a BPR loss.
- **MultiModalNCF**: our multi-modal extension of NeuMF. A content tower projects each modality (text 384-d, image 2,048-d, genre 18-d) to 32 dimensions each, concatenates them to 96 dimensions, then projects to 32 dimensions before fusing with the behavioural embedding.
- **Positive threshold**: neural models binarise ratings ≥ 4 as positive during training.
- **Negative sampling**: four negatives per positive for NeuMF and MultiModalNCF; BPR-based models use standard BPR sampling.

3.4 Evaluation protocol

We use the standard leave-last-5-out split: the five most recent interactions of each user are held out as the test set. All models are evaluated against the same candidate set (all items minus the user's training interactions). We report HR@K and NDCG@K at $K \in \{10, 20\}$; Recall@10 is additionally plotted in Figure 4.

Neural models and both baselines are trained with five random seeds (42–46) and reported as mean $\pm$ standard deviation. Deterministic models (UserCF, ItemCF, SVD) have zero spread. Figure 3 summarises the protocol.

3.5 Implementation, reproducibility and engineering quality

The pipeline is implemented in Python; content features are precomputed once and shared by all content-aware models, and every experiment is driven by a configuration file so that a run is reproducible from its parameters alone. Experiments run on commodity hardware through the repository's `make` targets. To support reproducibility we maintain a well-tested codebase with continuous integration:

- **Test suite**: 125 tests passed, with 50.9% line coverage (minimum threshold 45%). Tests cover data loading, model training loops and evaluation metrics.
- **CI pipeline**: three jobs run on every push to main and every pull request—linting plus tests with coverage, end-to-end accessibility tests, and Docker image builds for both deployment modes (lite and full).
- **Dependency hygiene**: we pin version lower bounds to ensure reproducible environments. Initial vulnerability scans flagged known advisories in transitive dependencies; after updating to patched versions, no known vulnerabilities remain (as of October 2026).
- **Documentation**: key architectural decisions are recorded in Architecture Decision Records (ADRs), and the codebase provides `make` commands for setup, data preparation, training and evaluation, allowing full reproduction from scratch.
- **Repository hygiene**: pre-commit hooks reject large files, malformed YAML and merge-conflict markers before each commit, and the tracked text files contain no credentials or secret material.

4. Results

All experimental results below are read from the committed result artefacts of release v1.0.0 (2026-10-13) and are reproducible via the repository's `make` targets; the continuous-integration figures are as reported by the CI run on v1.0.0.

4.1 Main results

Table 2 shows the main model comparison on MovieLens 100K. Key observations:

- The expected ordering UserCF < ItemCF < SVD holds, with the neural band (0.285–0.371 HR@10) sitting above SVD.
- BPR-MF and NeuMF are within one combined standard deviation of each other.
- VBPR's HR@10 edge over MultiModalNCF (0.0055) is smaller than one MultiModalNCF standard deviation, while MultiModalNCF leads both NDCG columns; the pair is therefore read as statistically tied.

Figure 4 visualises the same comparison at K=10.

4.2 Modality ablation

Table 3 shows the modality ablation study on MultiModalNCF. Removing all content lowers every metric, but by roughly one seed standard deviation. The single-modality deltas sit inside the noise (one even reverses sign), so no per-modality ranking is claimed. Figure 5 plots the per-variant HR@10 and NDCG@10.

4.3 Cold-start study

Table 4 shows the cold-start experiment, in which 100 items are removed from training and evaluated as cold-start items. The ordering content > blind > random holds on every metric. The HR@10 spread of the content variant (0.0290) exceeds the gap to NeuMF (0.0193), so the margin is reported as a direction rather than a point magnitude. Figure 6 visualises the three-way comparison.

4.4 Cross-dataset validation

Table 5 compares performance across both datasets, and Table 6 repeats the modality ablation on Amazon.

What transfers: the ladder ordering, the BPR-MF ≈ NeuMF tie, the "single-modality deltas are inside seed noise" conclusion (on Amazon, removing any single modality moves HR@10 by at most 0.0030, still inside the 0.0085 full-model spread), and the cold-start direction—the content model reaches HR@10 0.2364±0.0202 against 0.1204±0.0120 for the content-blind model and 0.1092 for random, a content/blind ratio of 1.96× (1.17× on MovieLens). What does not: on Amazon the top HR@10 point estimate belongs to LightGCN (0.1187 vs 0.1164), a difference well inside the combined spread, and Behaviour-only ≈ Full (0.1163 vs 0.1164).

4.5 Training cost

Figure 7 reports the per-model training cost behind Table 2.

5. Discussion

5.1 Key findings

1. **Model ordering is largely consistent across domains.** The ladder backbone (UserCF < ItemCF < SVD) and the neural band's lead over SVD hold on both MovieLens and Amazon, suggesting that these patterns are not dataset-specific artefacts. Which model leads *within* the neural band is domain-dependent, however (VBPR/MultiModalNCF on MovieLens, LightGCN on Amazon), so no cross-domain "best model" is claimed.
2. **Content features help in cold-start, but not much in overall ranking.** The modality ablation shows that removing content features has limited impact on overall ranking quality, whereas the cold-start experiments show measurable gains when content is available.
3. **MultiModalNCF is competitive with established baselines.** Despite using a simple concatenation-based fusion, MultiModalNCF performs on par with VBPR, a well-established content-aware baseline.

5.2 Comparison with published baselines and related work

Our results are read against two baselines that we reproduce in-house rather than against their originally reported scores, because the split, the candidate set and the metric definitions differ across papers. Under this matched protocol the direction of VBPR's original claim [2] is reproduced: a content-aware model is clearly stronger than a content-blind model on cold-start items. The same comparison also bounds that claim—on these two datasets the content channels contribute little to overall ranking quality, so the advantage of a multi-modal model over a strong content-blind model such as NeuMF is not established. This is consistent with the wide-spread observation that visual and textual side information mainly compensates for missing interaction history.

5.3 Limitations

1. **Single split.** We use one leave-last-5-out partition by design; conclusions are read against seed spread, not split variance, and no cross-validation is performed.
2. **No significance testing.** Differences are compared against reported standard deviations; formal statistical tests are not run. Small deltas (e.g. VBPR vs MultiModalNCF HR@10) are explicitly left undecided.
3. **Epoch budgets are not equalised.** MultiModalNCF runs 10 epochs against VBPR's 30 and BPR-MF's/LightGCN's 50, so accuracy comparisons carry that asymmetry.

4. **Training-time measurement conditions.** Hardware and warm/cold cache conditions are not recorded alongside the reported seconds.
5. **Second-domain slice represents heavy users only.** The Amazon slice is restricted to users with ≥ 20 ratings, chosen to match the MovieLens population rule, and does not describe the full Amazon population.
6. **Content ablation effect is small.** Removing all content costs about one seed standard deviation on MovieLens and nothing measurable on Amazon.
7. **Free-tier deployments sleep when idle.** Reported live behaviour may include cold-start latency on the first request.

6. Conclusion

We presented CineRec, a practical comparative study of six recommendation models across two domains, with two published baselines and multi-seed validation. Our findings confirm that (1) model rankings are largely consistent across datasets, (2) content features provide measurable benefits in cold-start scenarios but little overall ranking benefit, and (3) a simple multi-modal extension of NeuMF can compete with an established content-aware baseline such as VBPR.

Future work includes (1) evaluating on larger-scale datasets such as KuaiRec or Yelp, (2) exploring more sophisticated fusion strategies beyond simple concatenation, (3) conducting formal significance testing, and (4) extending the cold-start study from item cold-start to user cold-start. All code and evaluation scripts are available at `https://github.com/ElijahZhao/cinerec`.

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AI use disclosure: This project was assisted by AI tools throughout development and manuscript preparation, including code generation, debugging and documentation writing. The author is solely responsible for the study design, the verification of all reported results, and the content of this paper.

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Figure Legends

Figure 1. Overview of the CineRec study. Two domains feed precomputed text/image/genre content features into a ladder of eight recommenders—six ladder models plus two reproduced published baselines (marked *)—all evaluated under a single leave-last-5-out protocol with five-seed mean±std reporting.

Figure 2. MultiModalNCF architecture (the configuration used in all reported runs). Behavioural path: GMF over user/item embeddings (d=32). Content tower: text/image/genre are each projected to 32 dimensions, concatenated to 96, then projected to a 32-dimensional content embedding; the MLP path replaces the item embedding with this content embedding. The two paths are concatenated (64) and mapped to a score. In the cold-start path the GMF item vector is zeroed, so only the content tower drives scoring.

Figure 3. Evaluation protocol. A per-user leave-last-5-out split defines training and held-out interactions; held-out items are ranked against a candidate set of all non-training items; HR@K and NDCG@K ($K \in \{10, 20\}$) are reported over five seeds (42–46) as mean±std. The protocol is run independently on both domains.

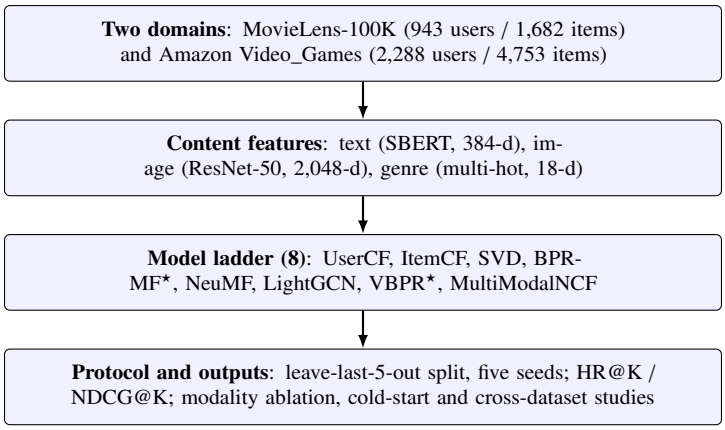
Figure 4. Model comparison on MovieLens 100K at K=10 (HR, NDCG, Recall). The HR@10 and NDCG@10 bars are the five-seed means from Table 2; Recall@10 is from the same evaluation artefacts.

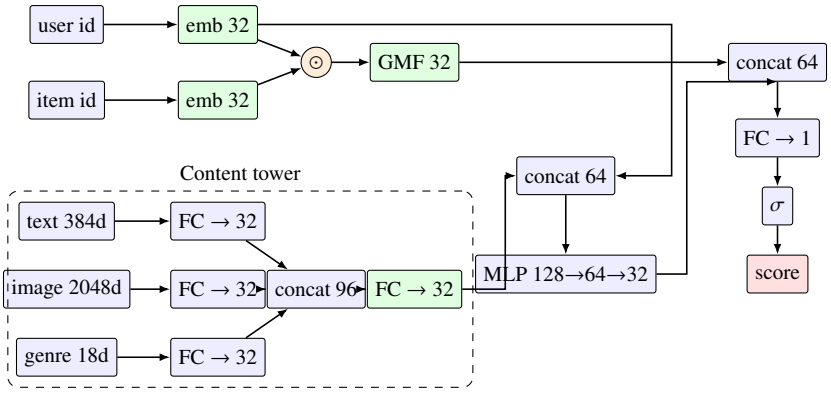
Figure 5. Modality ablation on MultiModalNCF (MovieLens 100K): HR@10 and NDCG@10 for the full model and each ablated variant. The values match Table 3.

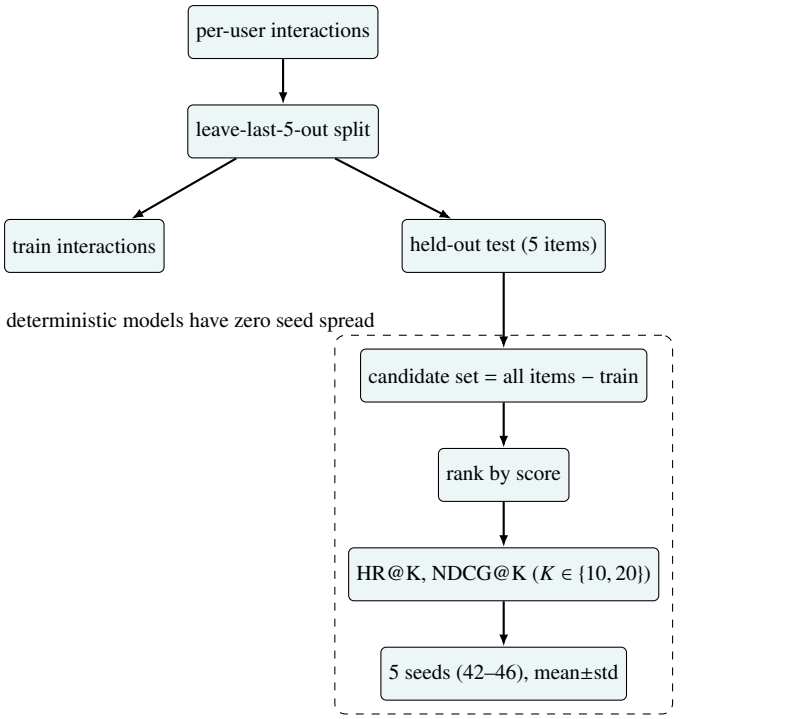
Figure 6. Cold-start study: 100 items held out of training and serving as the only candidates, comparing a content-aware model, a content-blind model and a random ranker. The values match Table 4.

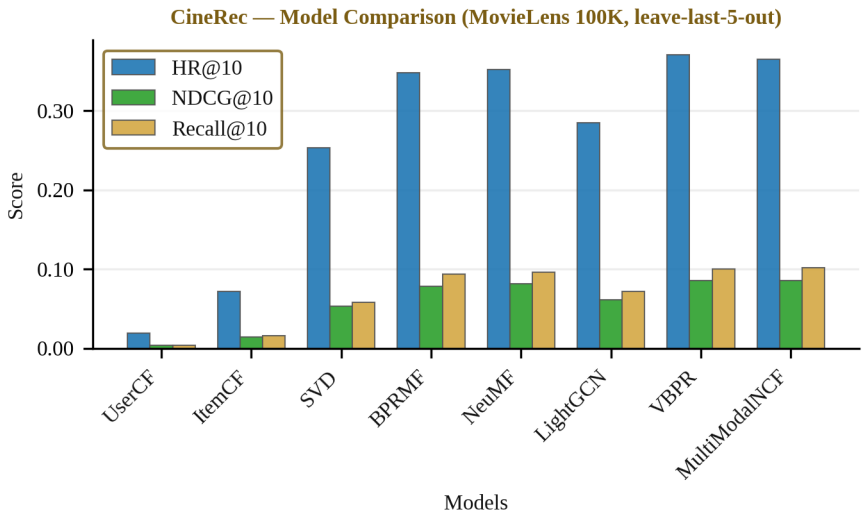
Figure 7. Training time per model (five-seed means, same runs as Table 2). Measurement conditions are not recorded; see the limitations in Section 5.3.

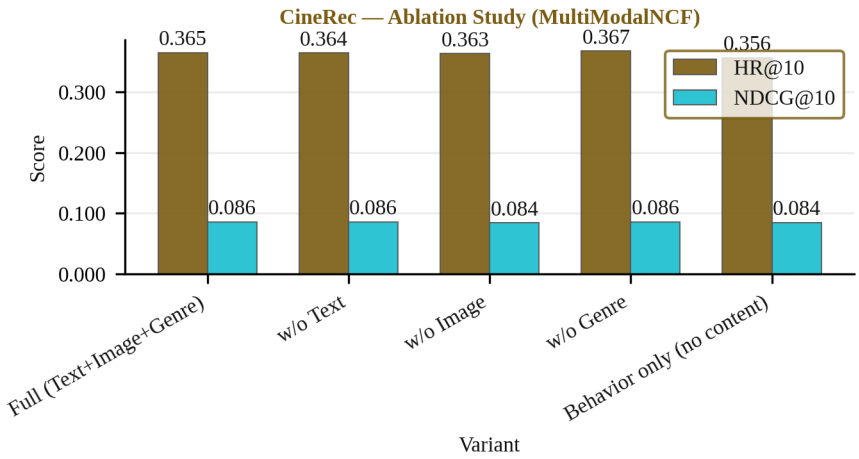
Figures

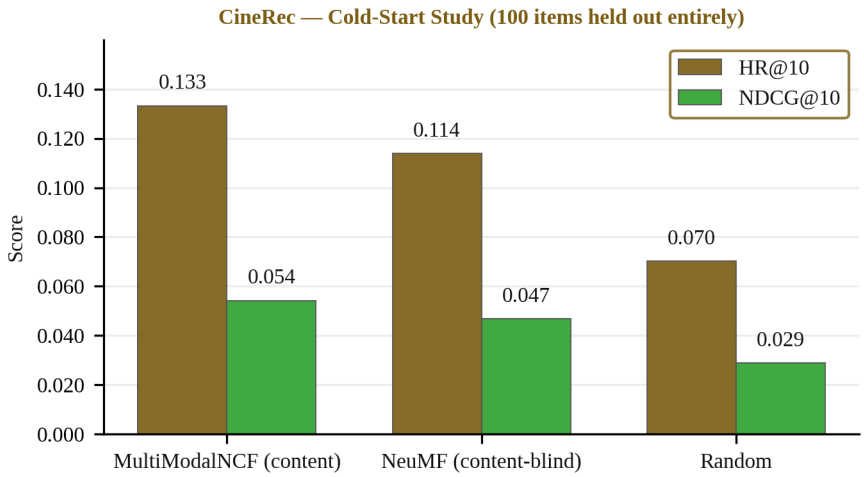

Two domains: MovieLens-100K (943 users / 1,682 items) and Amazon Video_Games (2,288 users / 4,753 items)
Content features: text (SBERT, 384-d), image (ResNet-50, 2,048-d), genre (multi-hot, 18-d)
Model ladder (8): UserCF, ItemCF, SVD, BPR-MF⋆, NeuMF, LightGCN, VBPR⋆, MultiModalNCF
Protocol and outputs: leave-last-5-out split, five seeds; HR@K / NDCG@K; modality ablation, cold-start and cross-dataset studies

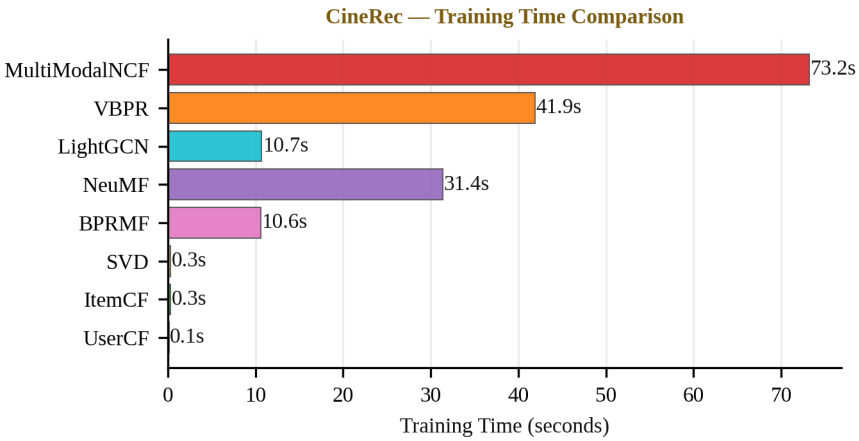

user id
emb 32
item id
emb 32
GMF 32
concat 64
FC → 1
σ
score
concat 64
MLP 128→64→32
Content tower
text 384d
FC → 32
image 2048d
FC → 32
concat 96
FC → 32
genre 18d
FC → 32


per-user interactions
leave-last-5-out split
train interactions
held-out test (5 items)
deterministic models have zero seed spread
candidate set = all items − train
rank by score
HR@K, NDCG@K ($K \in \{10, 20\}$)
5 seeds (42–46), mean±std
run independently on MovieLens-100K and Amazon Video_Games


CineRec — Model Comparison (MovieLens 100K, leave-last-5-out)
HR@10
NDCG@10
Recall@10
Score
0.00
0.10
0.20
0.30
UserCF
ItemCF
SVD
BPRMF
NeuMF
LightGCN
VBPR
MultiModalNCF
Models


CineRec — Ablation Study (MultiModalNCF)
HR@10
NDCG@10
Score
0.000
0.100
0.200
0.300
0.365
0.364
0.363
0.367
0.356
0.086
0.086
0.084
0.086
0.084
Full (Text+Image+Genre)
w/o Text
w/o Image
w/o Genre
Behavior only (no content)
Variant


CineRec — Cold-Start Study (100 items held out entirely)
HR@10
NDCG@10
Score
0.000
0.020
0.040
0.060
0.080
0.100
0.120
0.140
0.133
0.054
0.114
0.047
0.070
0.029
MultiModalNCF (content)
NeuMF (content-blind)
Random


CineRec — Training Time Comparison
MultiModalNCF
VBPR
LightGCN
NeuMF
BPRMF
SVD
ItemCF
UserCF
73.2s
41.9s
10.7s
31.4s
10.6s
0.3s
0.3s
0.1s
0
10
20
30
40
50
60
70
Training Time (seconds)

Tables

Table 1. Model configurations.

Model	Configuration
UserCF	Pearson similarity, $k = 50$ neighbours
ItemCF	Adjusted cosine similarity, $k = 50$ neighbours
SVD	Truncated SVD, $k = 64$ latent dimensions
BPR-MF	dim 64, epochs 50, lr 0.001, batch 4,096, patience 8
NeuMF	dim 32, MLP (128, 64, 32), epochs 10, batch 512, four negatives, patience 3
LightGCN	dim 64, three layers, epochs 50, batch 4,096, BPR objective, patience 8
VBPR	behaviour dim 64, visual dim 64, epochs 30, lr 0.005
MultiModalNCF	dim 32, MLP (128, 64, 32), epochs 10, batch 512; content tower projects 384/2,048/18 $\rightarrow$ 32 each, concatenated (96) $\rightarrow$ 32

Table 2. Model comparison on MovieLens 100K (mean±std over five seeds; training time in seconds).

Model	HR@10	NDCG@10	HR@20	NDCG@20	Train time (s)
UserCF	0.0191±0.0000	0.0035±0.0000	0.0477±0.0000	0.0060±0.0000	0.1
ItemCF	0.0721±0.0000	0.0142±0.0000	0.1283±0.0000	0.0199±0.0000	0.3
SVD	0.2534±0.0000	0.0532±0.0000	0.3542±0.0000	0.0664±0.0000	0.3
BPR-MF	0.3476±0.0116	0.0785±0.0018	0.4831±0.0116	0.1034±0.0018	10.6
NeuMF	0.3516±0.0036	0.0814±0.0016	0.4978±0.0073	0.1083±0.0017	31.4
LightGCN	0.2850±0.0041	0.0614±0.0004	0.4250±0.0047	0.0816±0.0007	10.7
VBPR	0.3705±0.0048	0.0855±0.0009	0.5200±0.0096	0.1130±0.0008	41.9
MultiModalNCF	0.3650±0.0077	0.0859±0.0017	0.5073±0.0148	0.1144±0.0015	73.2

Table 3. Modality ablation on MultiModalNCF (MovieLens 100K).

Variant	HR@10	NDCG@10	HR@20	NDCG@20
Full	0.3650±0.0077	0.0859±0.0017	0.5073±0.0148	0.1144±0.0015
w/o Text	0.3642±0.0100	0.0855±0.0024	0.5075±0.0108	0.1144±0.0025
w/o Image	0.3631±0.0085	0.0842±0.0026	0.5080±0.0029	0.1132±0.0025
w/o Genre	0.3673±0.0080	0.0857±0.0015	0.5107±0.0139	0.1144±0.0016
Behaviour only	0.3565±0.0061	0.0844±0.0024	0.4931±0.0089	0.1123±0.0033

Table 4. Cold-start performance (100 items removed from training).

Model	HR@10	NDCG@10	HR@20	NDCG@20
MultiModalNCF (content)	0.1333±0.0290	0.0542±0.0151	0.2754±0.0233	0.0882±0.0139
NeuMF (content-blind)	0.1140±0.0147	0.0467±0.0069	0.2351±0.0129	0.0739±0.0038
Random	0.0702	0.0288	0.2018	0.0610

Table 5. Cross-dataset validation: HR@10 / NDCG@10 comparison.

Model	ML-100K HR@10	Amazon HR@10	ML-100K NDCG@10	Amazon NDCG@10
UserCF	0.0191	0.0122	0.0035	0.0017
ItemCF	0.0721	0.0214	0.0142	0.0035
SVD	0.2534	0.0651	0.0532	0.0116
BPR-MF	0.3476	0.1055	0.0785	0.0198
NeuMF	0.3516	0.1053	0.0814	0.0190
LightGCN	0.2850	0.1187	0.0614	0.0222
VBPR	0.3705	0.1097	0.0855	0.0196
MultiModalNCF	0.3650	0.1164	0.0859	0.0213

Table 6. Modality ablation on Amazon Video_Games (MultiModalNCF, HR@10, mean±std over five seeds).

Variant	HR@10
Full	0.1164±0.0085
w/o Text	0.1159±0.0050
w/o Image	0.1177±0.0033
w/o Genre	0.1194±0.0017
Behaviour only	0.1163±0.0011